

Effects of high-temperature treatment on *in vitro* protein digestibility and protein profiles of pea ingredients obtained from wet and dry fractionation

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ABSTRACT

Thermal treatment can improve plant protein digestibility by inactivating antinutrients and partially denaturing proteins, but depending on the conditions applied, heating may also promote aggregation and amino acid modifications that hinder protein digestibility. This study describes how heat affects the protein profile and *in vitro* digestibility of pea ingredients produced by wet and dry fractionation. Protein suspensions (40 wt%) of pea flour (PF), protein-rich fraction (PRF), and protein isolate (PI) were heated at 90 °C or 140 °C for 20 min. Proteomics, SDS-PAGE, trypsin inhibitor activity (TIA), and *in vitro* protein digestion (INFOGEST) were performed. All heated samples showed protein aggregation, likely limiting protein identification in PF and PRF. The highest temperature increased protein oxidation in PI and markedly lowered TIA in PRF (87%), improving its intestinal digestion. Nevertheless, heating did not alter overall *in vitro* digestibility, suggesting that matrix effects offset the advantages of TIA reduction.

1. Introduction

Plant-based products are considered a way to mitigate the impact of the food system on climate change. Especially plant protein-rich products play a role in helping consumers reduce their meat intake. These products have grown significantly over the past decade (GFI, 2025), yet significant research gaps remain in understanding their nutritional and sustainability benefits.

Among plant protein sources, pea seeds (*Pisum sativum*) are widely used in a diverse range of plant-based foods and often serve as an alternative to soybeans (Grossmann, 2024). Peas are also an attractive crop for low-input farming because their nitrogen-fixing symbiosis reduces fertilizer requirements and adds nitrogen to the soil when used in crop rotations (Winther et al., 2023). Beyond these favourable cultivation characteristics, peas are relevant to the food industry because of their macronutrient profile: on a dry basis, they contain 39–46% starch, 23–30% dietary fiber, and 20–25% protein, consisting primarily of globulins and, to a lesser extent, albumins. Pea proteins are typically rich in lysine, while being limited in sulfur-containing amino acids (Wu

et al., 2023). However, like other legumes, peas also contain several antinutrients, such as protease inhibitors that impact protein nutritional quality. These inhibitors are known to limit protein digestion due to the inactivation of the digestive enzymes trypsin and chymotrypsin in the intestinal tract (Kårlund et al., 2021). The two protease inhibitors present in legume seeds are Kunitz trypsin inhibitor (KTI) and Bowman-Birk protease inhibitor (BBI). KTI primarily inhibits trypsin, while BBI inhibits both trypsin and chymotrypsin (Bendre et al., 2018; Gu et al., 2014). BBI is particularly stable, due to the presence of the seven intrachain disulfide bonds, and can maintain inhibitory activity even after exposure to high temperatures and across a broad pH range (Gitlin-Domagalska et al., 2020). Therefore, food processing is crucial to reduce or inactivate protease inhibitors in legumes, because they hinder protein digestibility. Among the processes used in the industry, heat treatments and wet fractionation processes are efficient in inactivating/removing (part of) protease inhibitors in a variety of plant protein sources (Amin et al., 2022).

Currently, plant proteins are typically extracted through conventional wet fractionation processes, such as alkaline or acid treatments to

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solubilize proteins, followed by isoelectric precipitation and drying. Wet fractionation yields a purified protein ingredient (i.e., protein isolate, PI) with high protein content, containing mainly the globulin fraction (Bessada et al., 2019). Because of their high purity, the proteins in PI are expected to have high *in vitro* protein digestibility, since the proteins are partially denatured and most of the antinutrients have been removed or inactivated during the fractionation process (Amin et al., 2022). On the other side, protein aggregation is often observed in these ingredients, especially as a result of the drying step, which is the final stage in the process (Yang et al., 2022). Beyond functional and nutritional considerations, wet fractionation generally involves greater water and energy inputs than dry fractionation, resulting in a comparatively lower sustainability performance (Schutyser et al., 2025; Vogelsang-O'Dwyer et al., 2020). A holistic assessment that integrates both environmental impact and nutritional quality is therefore important.

During dry fractionation, starch granules and smaller protein-rich particles are usually separated from the milled flour with air classification based on size and density differences. Dry fractionation gives a starch-rich fraction and a protein-rich fraction (PRF), with lower purity than PI, containing both globulin and albumin fractions, fibers, antinutrients, as well as some remaining starch. The presence of antinutrients in the PRF can represent a disadvantage of dry fractionation and could result in lower protein digestibility compared to PI (Vogelsang-O'Dwyer et al., 2020). Therefore, depending on the fractionation route, the protein ingredient will present a different protein profile, structure, functional property, and digestibility (Grossmann, 2024). However, these initial differences may be further altered when the protein ingredients undergo additional processing steps before becoming a final plant-based food product.

The manufacturing of protein-rich plant-based foods involves subsequent processing of blends of ingredients that often use heating and mechanical stress to achieve the desired food texture (e.g., extrusion). Depending on the heating conditions applied during processing, thermal denaturation either improves or hinders protein digestibility. Partially unfolded proteins expose proteolytic sites to digestive enzymes, thereby enhancing digestion, whereas protein aggregation reduces enzyme accessibility, resulting in the opposite effect (Ajomiwe et al., 2024; Orlén et al., 2023). Additionally, thermal treatment can induce protein oxidation and Maillard reactions, leading to irreversible modifications of amino acids (Estévez et al., 2024; Poojary & Lund, 2022), which can affect their bioavailability (Ajomiwe et al., 2024). Because additional processing is required to transform protein ingredients into food products, an important question is whether the improved nutritional quality often attributed to more extensively processed, and less sustainable, ingredients such as PI is maintained after further heat treatments. The fractionation routes not only impose different processing histories, resulting in distinct protein structures and profiles, but also produce ingredients with markedly different matrix compositions. These differences are expected to influence how proteins respond to further food processing and digestion, thereby ultimately affecting their nutritional quality.

The aim of this study was to quantify how heat treatment affects the protein profile and *in vitro* digestibility of pea flour and pea protein ingredients obtained by dry (PRF) and wet fractionation (PI). Protein structural and oxidation modifications were evaluated using SDS-PAGE and proteomics, while nutritional properties were assessed by trypsin inhibitor activity and *in vitro* protein digestibility using the INFOGEST method. We hypothesized that high-temperature treatment induces more extensive protein modifications, including aggregation and oxidation, and as a consequence limits protein digestibility under *in vitro* conditions, despite the expected trypsin inhibitor inactivation.

2. Material and methods

2.1. Materials

Dehulled yellow peas (*Pisum Sativum* L.) and PI were kindly provided by Biopimate Aps (DK) and stored in a closed container in the dark at room temperature. All protein ingredients, including PI, were produced from the same batch of yellow peas. All materials and chemicals (analytical grade) were obtained from Sigma-Aldrich Denmark A/S (Søborg, DK). All solutions were prepared using ultrapure water (Milli-Q water system, Millipore Corporation, Merck), unless stated otherwise.

2.2. Preparation of pea ingredients

Pea flour (PF) was obtained by milling dehulled peas into a fine flour with a ZPS50 impact mill (Hosokawa-Alpine, Augsburg, DE) under the following conditions: mill speed of 8000 rpm, an airflow of 52 m³/h, a classifier wheel speed of 5000 rpm and a feed rate of 10 rpm, according to (Pelgrom et al., 2013). After milling, PF was dry fractionated to obtain a PRF using air classification (ATP50 Hosokawa-Alpine, Augsburg, DE) with a classifier wheel speed of 5000 rpm and similar settings for the screw feeder rate and airflow as during the milling. After dry fractionation, the pea PRF was stored in the dark in a closed container at room temperature until further analysis. The starch-rich fraction was not used for further analysis. The PI was produced by aqueous acid extraction followed by isoelectric precipitation, called here as wet fractionation (Markedal et al., 2012).

2.3. Thermal treatment of protein ingredients

To investigate the effect of thermal treatment on the pea ingredients, two temperatures were chosen, low and high, to simulate mild and harsher conditions that are used to make plant-based products. The conditions applied for the heat treatment were according to those applied in the High Temperature Shear Cell (HTSC) conditions, such as dry matter content, the highest temperature, and time (Schreuders et al., 2019). All samples (40 wt%) were first manually mixed with demi-water and then stirred at 150 rpm using a magnetic stirring plate for 1 h to allow hydration. Then, samples were loaded into metal capsules (5 cm diameter) and properly sealed before being heated at 90 °C (water bath) and at 140 °C (oil bath) for 20 min. After heating, the capsules were cooled down in a water-iced bath for 5 min. A control sample was prepared under the same conditions without the heat treatment, which will be referred to here as an unheated sample. Samples were stored at −20 °C and freeze-dried for 72 h using a Christ Epsilon 2-6D freeze-dryer (Martin Christ Gefriertrocknungsanlagen GmbH, Osterode am Harz, DE). After freeze-drying, the samples were stored in air-tight containers at room temperature until analysis. Before analysis, samples were sieved through a 0.3 mm sieve (Buch & Holm A/S, Herlev, DK), and the entire sample was forced through the sieve.

2.4. Chemical composition

The total nitrogen content of all samples was determined by combustion using an Elementar Vario MACRO Cube CHNS Analyzer (Elementar Analysensysteme GmbH, Hanau, DE), and total crude protein content was calculated using a conversion factor of 6.25. Samples were analyzed in triplicate, and aspartic acid was used as an internal standard. Crude fat was determined in triplicate by the Soxhlet method using a rapid and automated extraction system, Soxtherm (Gerhardt GmbH & Co, Königswinter, DE) with petroleum ether as a solvent, and an extraction temperature of 150 °C for 3 h. The moisture content of protein ingredients was determined in an oven at 105 °C overnight, in duplicate. Starch content was analyzed by an external accredited laboratory of SGS Germany GmbH (Hamburg, DE) on single replicates.

2.5. Scanning Electron microscopy

Scanning electron microscopy (SEM) images were obtained of the pea ingredients deposited on carbon conductive tape on aluminum SEM stubs, analyzed using a Phenom G2 Pure Senseo microscope (Thermo Fisher Scientific, Inc.). The samples were analyzed at 5.0 kV, at room temperature, at a working distance of 13.4 mm, and at $\times 1000$ magnification.

2.6. Total phenolic content

The total phenolic content (TPC) of the pea ingredients was determined following the method described by Singleton et al. (1999) with some modifications. Briefly, 2 g of pea ingredients were dispersed in 10 mL methanol solution (40%), vortexed for 1 min, and centrifuged at $2000 \times g$ for 10 min. The supernatant was taken, and the extraction in methanol was repeated once more. Afterward, the supernatants were diluted up to 25 mL in ultrapure water. A standard curve using gallic acid (0–50 $\mu\text{g/mL}$) was prepared by first preparing a stock solution with 25 mg of gallic acid homogenized with 5 mL ethanol (99%) and further diluted to 50 mL in ultrapure water. For the assay, 100 μL of sample, standards, or water blank were diluted with 700 μL of ultrapure water in Eppendorf tubes and mixed with 50 μL of Folin-Ciocalteu reagent. After 30 s, 150 μL sodium carbonate (20%) was added, and samples were vortexed for 5 s. Then, samples were incubated at 40°C for 30 min in a water bath. After incubation time, 300 μL of the sample was transferred to a 96-well microplate, and the absorbance was measured at 765 nm using a microplate reader, Biotek Epoch 2 Microplate Reader (Agilent, Santa Clara, USA). The same procedure was done for the standard curve and blank. TPC was expressed as milligrams of gallic acid equivalents (GAE) per gram of sample on a dry basis.

2.7. Protein molecular weight distribution

The protein molecular weight distribution of the unheated and heated protein ingredients was determined using Sodium-Dodecyl-Sulfate Polyacrylamide Gel Electrophoresis (SDS-PAGE), under reducing and non-reducing conditions. Protein extraction was carried out as follows: 10 mg/mL suspension (PI and PRF) and 20 mg/mL suspension (PF) were prepared in 0.1 M Tris buffer (containing 5% SDS, pH 8.0) and left for hydration overnight under rotation at 40 rpm using a rotating wheel (Rotator SB3, Bibby Sterilin Ltd., Stone, UK) at room temperature. The next day, samples were placed in a mixer mill (MM400, Retsch GmbH, Haan, DE) at 30 Hz frequency for 5 min, followed by an ultrasound bath at 47 kHz and 80 W (Branson 1210E-MT, Branson Ultrasonics Corporation, Danbury, USA) for 5 min. Then, samples were heated at 95°C under agitation at 350 rpm for 5 min in a shaking incubator (ProvoCell™ shaking micro incubator, Esco Life Sciences Division, Singapore, SG). Afterward, samples were cooled to room temperature and centrifuged at $20,000 \times g$ for 5 min. The supernatant was collected, and the soluble protein concentration was measured at 280 nm using a nanodrop UV-Vis spectrophotometer (Thermo Fisher Scientific, Waltham, MA). Determination of protein concentration at 280 nm is a simple and quick method; however, it may overestimate the protein content in the flours, as they contain other compounds that can also absorb at this wavelength. Then, 65 μL of the sample was diluted to a final protein concentration of 1 mg/mL in a mixture containing 25 μL of $4\times$ LDS sample buffer (lithium dodecyl sulfate, pH 8.4) and 10 μL of 1 M dithiothreitol (DTT). For the non-reduced samples, ultrapure water in the same volume was used instead of DTT. Samples were then heated at 80°C and 350 rpm for 10 min in a shaking incubator. Before loading samples into the gels, the samples were centrifuged at $20,000 \times g$ for 5 min. Electrophoresis was carried out using 4–12% NuPAGE Bis-Tris gels (Thermo Fisher Scientific, Waltham, MA) and NuPAGE MOPS SDS running buffer ($20\times$) (Thermo Fisher Scientific, Waltham, MA) in an XCell Sure Lock Mini-Cell Electrophoresis System (Thermo Fisher

Scientific, Waltham, MA). The wells were loaded with 10 μL of sample and 3 μL of Novex™ Sharp Pre-stained protein standard (LC-5800, Thermo Fisher Scientific, Waltham, MA). Gels were run under 200 V for approximately 40 min. After running the gels, they were stained with 2% Coomassie Brilliant Blue and equilibration buffer (0.4 M ammonium sulfate, 0.2 M 86% phosphoric acid, 3.9 M 96% ethanol) in a ratio of 1:100 overnight on a platform shaker (Polymax 1040, Heidolph Scientific Products GmbH, Schwabach, DE). The following day, the gels were washed in ultrapure water and scanned using Epson Perfection V850 Pro scanner (Seiko Epson Corp., Nagano, JPN) and the software SilverFast 8 (LaserSoft Imaging AG, Kiel, DE).

2.8. Proteomics profile

Bottom-up proteomics was applied to all samples to determine the effect of thermal treatment on the profile of proteins and protein oxidation markers, according to Zhang et al. (2024). Briefly, sieved samples were solubilized (10 mg/mL) in 0.1 M Tris-HCl buffer (pH 8.0) containing 8 M urea overnight at room temperature under constant shaking using a platform shaker (Polymax 1040, Heidolph Scientific Products GmbH, Schwabach, DE). The next day, soluble protein concentration was determined by the BCA method at 562 nm using bovine serum albumin as a standard, according to the manufacturer's instructions (Thermo Scientific™ Pierce™ BCA Protein Assay Kit). The protein concentration was adjusted to 1.0 mg/mL and 100 μL of the samples were first diluted with 100 μL 1 M Tris-HCl buffer (pH 8.0) containing 8 M. Then, samples were reduced with freshly prepared 20 μL 450 mM DTT in 100 mM ammonium hydrogen carbonate (NH_4HCO_3) for 45 min at room temperature. Afterward, samples were alkylated with freshly prepared 40 μL 500 mM iodoacetamide solution in 100 mM NH_4HCO_3 for 1 h at room temperature in the dark. After incubation, 740 μL of 10 mM NH_4HCO_3 was added to the samples. Finally, samples were digested with 30 μL of 0.1 $\mu\text{g}/\mu\text{L}$ trypsin solution (recombinant bovine trypsin expressed in corn TrypZean®, 3350 units/mg solids, Sigma-Aldrich, Søborg, DK) overnight at 37°C and 700 rpm in a shaking incubator. The next day, the digest was acidified with 20 μL of 10% (v/v) trifluoroacetic acid, centrifuged at $20,000 \times g$ for 2 min, and the supernatant was transferred to a new tube. The addition of 10% trifluoroacetic acid was repeated twice on the pellet. Filtered through a 0.22 μm (membrane size) regenerated cellulose membrane syringe (Q-Max RR syringe filters, Frisette ApS, Knebel, DK). The filtrate was stored at -20°C until LC-MS/MS analysis.

The tryptic peptides were analyzed using an Orbitrap Exploris 480 mass spectrometer coupled to a Vanquish UHPLC system (Thermo Fisher, Roskilde, DK). Peptide separation was achieved using a bioZen™ Peptide XB-C18 column (1.7 μm particle size, 150×2.1 mm, Phenomenex, Værløse, DK) over a 40-min linear gradient of solvent A (0.1% formic acid) and solvent B (0.1% formic acid/80% acetonitrile). The mass spectrometer operated in positive ion mode with a full MS scan resolution of 60,000, a normalized automatic gain control (AGC) target of 300%, and a mass scan range of 200–2000 m/z . The 10 most intense spectra were selected for MS/MS analysis using an Orbitrap resolution of 30,000, higher-energy collisional dissociation (HCD) with 30% collision energy, a normalized AGC target of 100%, a 2.0 m/z isolation window, and a maximum injection time of 200 ms.

Proteomic data were processed using Proteome Discoverer software (version 2.5, Thermo Fisher Scientific, Waltham, USA) and searched against the *Pisum sativum* Uniprot database (proteome ID: UP001058974). A reduced database was created based on the top 200 proteins identified in a larger database. The protein sequences that were more than 95% identical are represented by one master sequence, and the sequence with the highest annotation level was selected (raw and processed data are available in the supplementary material 1). The search parameters included trypsin as the enzyme, a peptide length of 7–30 amino acids, a maximum of 2 missed cleavages, a precursor mass tolerance of 10 ppm, and a fragment mass tolerance of 0.07 Da. The

static modification cysteine carbamidomethylation (+57.021 Da) was considered, and the following dynamic modifications of monooxidation (+15.99 Da), dioxidation (+31.99 Da), and Lys to α -amino adipic semialdehyde (AAS) (−1.03 Da) were considered for analysis (raw data with all identified peptides and a list with the modified peptides used to determine total protein oxidation per sample type is shown in the supplementary material 2). The identified proteins were filtered by the false discovery rate (FDR) below 1% and with at least one matched unique peptide. For analysis, we included proteins that had high FDR and were annotated. Duplicate accessions with equal or higher similarity of 95% were not included. Results are shown as the mean abundance of three independent replicates, which do not provide an absolute concentration but refer to the relative quantity of proteins and peptides detected from the intensity of the MS signals, allowing for comparison of protein levels across different conditions.

2.9. Determination of the trypsin inhibitor activity

The trypsin inhibitor activity (TIA) was determined according to the American Oil Society (AOCS) Method Ba 12a-2020 (Liu et al., 2021). In brief, sieved samples (0.3 mm) were mixed with 10 mM sodium hydroxide (NaOH) solution in a 1:50 w:v ratio and stirred at 600 rpm for 3 h in a magnetic stirrer. After extraction, the pH of the samples varied between 9 and 11. The resulting sample extract was then diluted with ultrapure water to achieve a 30–70% trypsin inhibition. Following this, 1 mL of the diluted samples was mixed with 2.5 mL of substrate solution containing 0.4 mg/mL N- α -benzoyl-dl-arginine-p-nitroanilide (dl-BAPA) in 50 mM Tris buffer (pH 8.2) containing 20 mM calcium chloride (CaCl₂), which was prewarmed at 37 °C in a water bath (LSB Aqua Pro, Grant Instruments, Cambridge, UK). Then, 1 mL of 20 μ g/mL trypsin solution prepared in 1 mM hydrochloric acid (HCl) containing 5 mM CaCl₂ was added to the sample with substrate, and incubated at 37 °C for 10 min. After that, 0.5 mL of acetic acid solution (30% v/v) was added to terminate the reaction. For the sample and reference blanks, acetic acid was added before the trypsin solution. Subsequently, the reaction mixtures were centrifuged at 1500 $\times$ g for 5 min. The absorbance of the supernatant was measured at 410 nm using a UV–Vis spectrophotometer (Shimadzu UV-1800; Cole-Parmer Instrument Company Ltd., UK) with a 1.0 cm light path. The TIA was quantified as trypsin inhibited units (TIU) per mg of sample (dry basis) and calculated using the following formula:

$$TIU/mg \text{ sample} = \frac{(A410R - A410RB) - (A410S - A410SB)}{0.02} \div (V_s) \times (C_s) \quad (1)$$

Where: A410R: reference reading, A410RB: reference blank reading, A410S: sample reading, A410SB: sample blank reading, V_s: 1 mL of diluted sample extract used for the assay, C_s: concentration of diluted sample extract in mg sample/mL. In this assay, one trypsin unit (TU) is arbitrarily defined as an increase of 0.02 absorbance at 410 nm at pH 8.2 and 37 °C, per 5 mL of the reaction mixture (Liu et al., 2021). The sample concentration was adjusted for the dilution factor. The resulting TIU per mg sample values were adjusted to a dry matter basis, accounting for the moisture content of the samples. To facilitate the comparison of samples with varying protein contents, TIU per mg sample (dry basis) values were normalized by the protein content (%). All measurements were conducted using three independent sample extracts, and each extract was assessed twice per assay.

2.10. In vitro protein digestibility (INFOGEST method)

The *in vitro* protein digestion was determined in unheated samples and samples heated at 140 °C using the static INFOGEST harmonized protocol (Egger et al., 2026; Sousa et al., 2023), with some modifications. Before protein hydrolysis, enzyme activities and bile concentration were determined according to the INFOGEST protocol (Minekus et al., 2014). The activity measured of porcine pepsin (P7012, Sigma

Aldrich) was 2628 U/mg, of trypsin activity in pancreatin was 7.3 U/mg (from porcine pancreas, 8 x USP, P7545, Sigma Aldrich), and of bile acid concentration was 1.94 mmol/g (bovine bile, B3883, Sigma Aldrich). Sieved samples were normalized based on their protein content for 0.04 g protein and solubilized with ultrapure water up to 1 g before digestion. Blank samples were prepared with 0.25 g of protein-free cookie consisting only of fat and carbohydrates (Sousa et al., 2020), which were solubilized in ultrapure water up to 1 g. Then, to simulate the oral phase, 0.8 mL simulated salivary fluid (1.25 $\times$, pH 7, prewarmed to 37 °C), 5 μ L of 0.3 M CaCl₂, and 0.195 mL of ultrapure water were added to the samples. After vortexing the samples, 1.6 mL of simulated gastric fluid (1.25 $\times$, prewarmed to 37 °C) was added. The pH was adjusted to 3 with 2 M HCl (volumes were determined in a prior pH test), after which 1 μ L of 0.3 M CaCl₂ and the remaining ultrapure water were added before the enzyme addition, bringing the total final volume to 4 mL. Finally, 0.1 mL of pepsin solution freshly prepared in ultrapure water (providing 2000 U/mL in the digesta) was added. After incubation of 120 min at 37 °C (Buch & Holm A/S, Herlev, Denmark) under 25 rpm rotation in a multi-rotator (PTR-35, 360° vertical rotator, Grant Instruments Ltd., Cambridge, UK), 1 mL sample was taken to determine the release of free amino groups in the gastric phase. Digestion was stopped by heating the samples at 100 °C in a block heater (SBH200D, Stuart, Bibby Sterilin Ltd., Stone, UK) for 10 min, followed by cooling down in an ice bath for 5 min, and centrifugation at 13,000 $\times$ g for 15 min at 4 °C. After centrifugation, the supernatant was transferred to a new tube and stored at −20 °C prior to analysis. The remaining digesta was mixed with 1.28 mL of simulated intestinal fluid (1.25 $\times$, prewarmed to 37 °C), and the pH was adjusted to 7 with 2 M NaOH (volumes were determined in a prior pH test). Subsequently, 8 μ L of 0.3 M CaCl₂ was added, and the remaining ultrapure water was added before the enzyme addition, bringing the total final volume to 6 mL. Afterward, 0.375 mL of bile solution (corresponding to 10 mmol/L in the final digesta) and 0.750 mL pancreatin solution (providing 25 U trypsin activity/mL in the final digesta) were added; both solutions were freshly prepared in simulated intestinal fluid. Samples were incubated for 120 min under the same conditions as described for the gastric phase. Pancreatin solution was prepared as described by Sousa et al. (2023) to improve enzyme solubilization. The solution was subjected to an ultrasonic bath and subsequently centrifuged to remove insoluble material. The amount of enzyme added was reduced to 25% in the assay. The enzyme and cookie concentrations in the blank were reduced to minimize background absorbance during free amino group analysis. Preliminary tests using 100%, 25%, and 10% pancreatin concentrations showed that 25% effectively lowered background noise without affecting the final concentration of released free amino groups (data not shown). After incubation, digestion was stopped as described earlier for the gastric phase. All samples and blanks were digested in triplicate on the same day.

2.11. Determination of free amino groups (OPA method)

The concentration of free amino groups released after *in vitro* protein digestion was determined in samples after 2 h and 4 h incubation using the o-phthalaldehyde (OPA) method as described by Sousa et al. (2023). Briefly, samples and a stock solution of L-glutamic acid (100 mmol/L in 0.5 mol/L perchloric acid) were diluted with 0.1 mol/L di-sodiumtetraborate-decahydrate solution (borate solution) before analysis. Then the OPA working solution was freshly prepared by mixing 0.1 mol/L borate solution (1:2 v/v), 10% w/w sodium dodecylsulfate solution (1:10 v/v), 40 g/L OPA in absolute ethanol (1:50 v/v), 200 g/L 2-mercapto-ethanesulfonic acid (1:50 v/v), and 100 g/L Tergitoltm (1:50 v/v) in ultrapure water. Then, 8 μ L of samples and standards were mixed with 232 μ L of OPA working solution in a 96-well microplate, the plate was shaken for 5 s, and incubated at 30 °C for 10 min. After incubation time, absorbance was measured at 340 nm using a microplate reader (Biotek Epoch 2 Microplate Reader, Agilent, Santa Clara, USA). The concentration of free amino groups was calculated using the standard

curve with glutamic acid (0–8 mmol/L). A protein-free control (blank digestion) was included to account for background measurements of the added digestive enzymes. The intestinal phase was calculated by subtracting the gastric phase from the total digestibility after 4 h of incubation.

2.12. Statistical analysis

Proteome profile, protein oxidation, and protease inhibitors were analyzed on a logarithmic scale in separate two-way ANOVAs with fixed effects of sample (PF, PRF, PI), thermal treatment (unheated, 90 °C, 140 °C), and their interaction. Release of free amino groups during *in vitro* digestion was analyzed on a logarithmic scale in a three-way ANOVA with fixed effects of sample (PF, PRF, PI), thermal treatment (unheated, 90 °C, 140 °C), phase (gastric, intestinal), and their interactions, and with a random effect of *in vitro* unit correlated across phase. And the total release of free amino groups across both phases was analyzed on the logarithmic scale in a 2-way ANOVA with fixed effects of sample and thermal treatment. The logarithmic scale was used as it resulted in statistically valid models according to residual plots and normal quantile plots of residuals. The fitted statistical models were used to perform *post hoc* comparisons across sample, *post hoc* comparisons across thermal treatments within samples (proteome profile, protein oxidation, protease inhibitors), and *post hoc* comparisons across combinations of sample and thermal treatment within phase (total release of free amino groups). All these comparisons were done using Tukey test, where we report statistical significances on a 5% level. Analysis was done in R v.4.4.2 using the packages lme4 v.1.1–35.5 and emmeans v.1.10.7.

3. Results & discussion

3.1. Effects of thermal treatment on protein profile

3.1.1. Chemical and visual characterization of pea ingredients

The chemical composition of the pea ingredients can be seen in Table 1. Both fractionation methods resulted in pea protein ingredients with enhanced protein and fat content compared to the PF. The dry fractionation method resulted in a PRF with 42% protein (on dry basis) and wet fractionation in a PI containing 84% protein (on dry basis). The protein content in the PF and PI extracted from yellow peas was in line with that previously reported (Li et al., 2024; Pelgrom et al., 2015). However, in our study, the protein content in the PRF was slightly lower, and the starch content was correspondingly higher. An increased presence of fibers is observed in the SEM images of PRF (Fig. 1A), suggesting a higher concentration of these components in this fraction, along with round starch granules surrounded by protein bodies. The morphology of PF and PRF ingredients is similar, as shown in the SEM images (Fig. 1A), while in the PI ingredient, there is a predominance of spherical and smooth shapes with only a few particles showing invaginations. These morphological characteristics are typical of spray-dried particles

Table 1
Chemical composition and total phenolic content (TPC) of pea ingredients.

Composition	Pea flour	Protein-rich fraction	Protein isolate
Protein*(%)	23.31 ± 0.30	42.23 ± 0.68	84.50 ± 0.50
Starch (%)	56.21	25.64	2.10
Fat (%)	1.14 ± 0.08	1.57 ± 0.32	5.29 ± 0.19
Moisture %	6.78 ± 0.04	6.38 ± 0.00	4.90 ± 0.00
TPC			
(mg GAE/100 g sample)	43.48 ± 1.56	62.00 ± 2.35	28.09 ± 1.34

Results are shown as mean ± standard deviation, except for starch in which results are shown from a single sample. Results are shown on a dry basis. *Nitrogen-to-protein conversion factor: 6.25. GAE: gallic acid equivalent.

(Muhoza et al., 2024). Lower protein purity in ingredients obtained from dry fractionation is expected when compared to protein isolates due to the limitations of the process in effectively separating smaller protein bodies from larger starch granules (Li et al., 2024). Besides, differences in the chemical composition of pea ingredients observed in the literature can be a result of the parameters chosen for dry fractionation, the method used for determining chemical composition, and the variety of pea flour used.

The TPC (Table 1) of the pea ingredients was determined to indicate any antioxidant potential of the ingredients. Among the three ingredients, the TPC was concentrated in the PRF, whereas in PI it was reduced by 35% compared to PF. As expected, the wet fractionation process eliminated most of the phenolic compounds present in the PF by solubilizing these compounds in the extraction water that is discarded during the process (Pedrosa et al., 2020). It is important to highlight that in this study, the pea seeds were dehulled before producing the ingredients, which might have contributed to a lower initial TPC in the PF compared to the literature (Younis et al., 2023). Likewise, Fernando and Manthey (2022) reported that wet fractionation removed 83% of the TPC in black bean protein isolate compared to the flour. Phenolic compounds can bind to proteins either reversibly through noncovalent bonds or irreversibly through covalent bonds, with protein–phenolic interactions being influenced by pH, temperature, and ionic strength (Kieserling et al., 2024). These protein–phenolic interactions can change the secondary and tertiary structure of proteins, altering the functionality and nutritional properties of the proteins (Fernando & Manthey, 2022). The colour changes observed in PF and PRF samples heated at 140 °C could indicate either protein–phenolic interactions or Maillard reaction, both of which are favoured at high temperatures (Fig. 1C) (Keppler et al., 2020). However, since pH is a key factor influencing protein–phenolic interactions, measuring the pH of the mixtures would have provided a basis to support this hypothesis. Given that the pH was not assessed and Maillard reaction products were not specifically investigated, these interpretations remain speculative, highlighting the need for further investigations to confirm the underlying mechanisms of colour changes. Additionally, the colour changes observed after heating might be a result of changes in other components present in the PF and PRF ingredients, considering the visual appearance of the samples.

3.1.2. Protein profile of pea ingredients

The molecular weight distributions of the proteins in the unheated and heated ingredients are shown in Fig. 1B. Under non-reducing and reducing conditions (Fig. 1B), the storage proteins convicilin (~70 kDa) and vicilin (~50 kDa) were identified in all samples, as expected (Husband et al., 2024). Legumin was identified between 60 kDa and 50 kDa when using non-reducing conditions (Fig. 1B.1). Under reducing conditions, both subunits of legumin were visible at ~40 kDa and ~20 kDa (Fig. 1B.2). PF and PRF had very similar protein profiles, which was expected since dry fractionation usually does not induce structural changes in the proteins nor removes part of the proteins. In the PI samples, it can be seen that fewer albumin proteins are present, as a result of the fractionation method. On top of the non-reduced gel, the formation of protein aggregates is visible (indicated by the arrow in the fig.) in the PI (unheated and heated at 90 °C and 140 °C) and in PRF and PF samples heated at 140 °C, while only low molecular weight proteins (<15 kDa) appear to be present. After reducing (Fig. 1B.2), the protein aggregates are still visible on top of the gel of samples heated at 140 °C, suggesting that these aggregates were stabilized by additional crosslinks in addition to disulfide bonds. Likewise, Jiménez-Munoz et al. (2023) showed that commercial pea PI unheated and heated at 90 °C for 15 min contained more large aggregates at the top of the gel than pea PRF (obtained by dry fractionation, both unheated and heated). Martineau-Côté et al. (2023) also observed an increase in bands above 250 kDa appearing at the top of the gel in samples of faba bean, peas, and soy flours after boiling compared to unheated samples. Tang et al. (2022) observed a similar pattern when heating pea protein isolate at 65 to

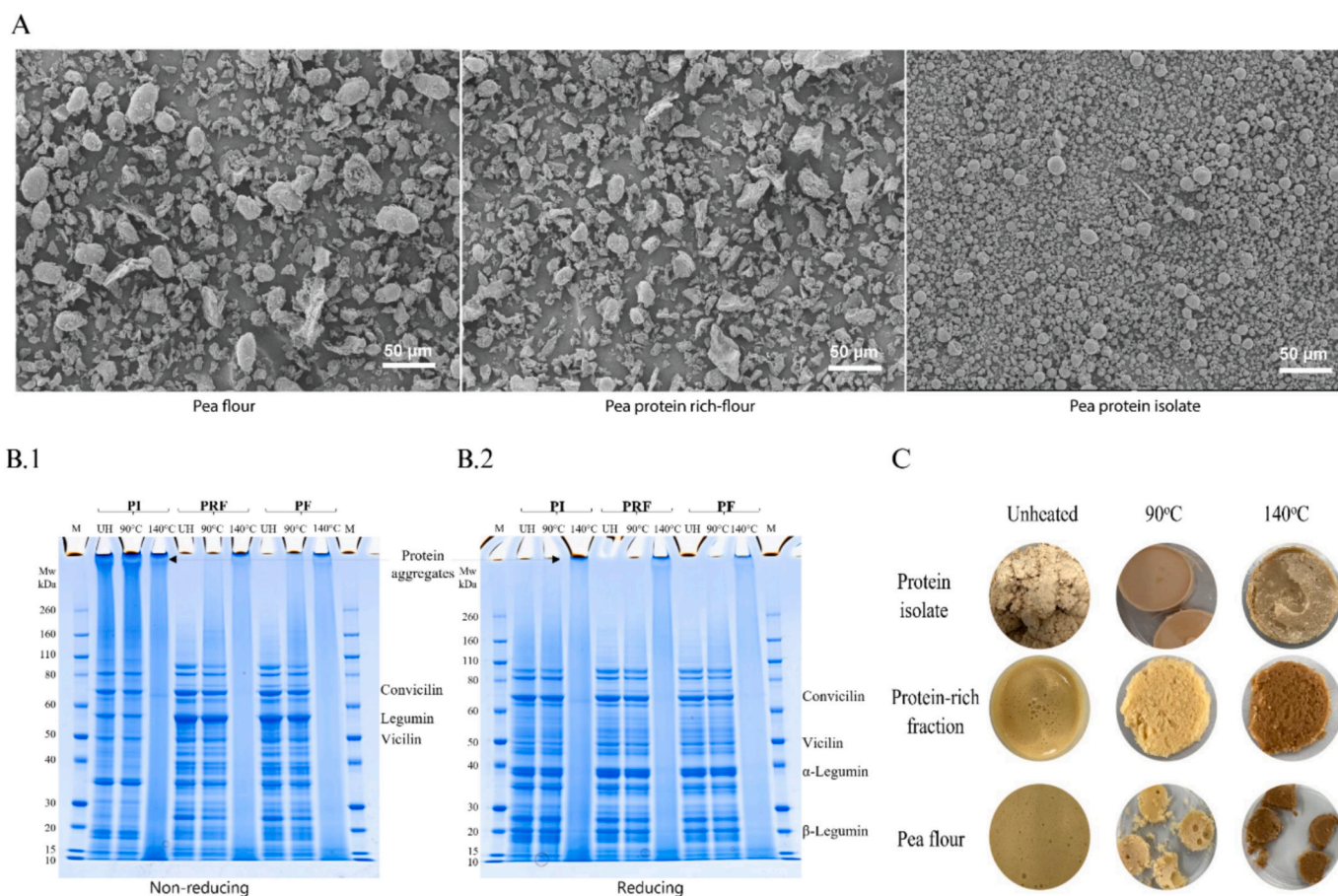


Fig. 1. A: Scanning electron micrographs of pea flour (PF), protein-rich fraction (PRF), and protein isolate (PI). The scale bar represents 50 μm . Magnification: $\times 1000$. B: Molecular weight distribution of protein ingredients obtained by Sodium Dodecylsulfate Polyacrylamide Gel Electrophoresis (SDS-PAGE), under non-reducing (B.1) and reducing (B.2) conditions. Mw: molecular weight; UH: unheated samples; M: molecular marker; C: Visualization of the appearance of the samples prepared before and after thermal treatment. Image tool: Adobe Illustrator (v. 27.0.1).

100 °C, where proteins already formed large soluble aggregates with molecular weights above 100 kDa at 65 °C. In contrast to our study, the authors did not find aggregates visible at the top of the wells under reducing conditions, indicating that the soluble aggregates were mainly stabilized by newly formed disulfide bonds. The presence of protein aggregates in the unheated PI may be attributed to the conditions applied during the wet fractionation (Verfaillie et al., 2023) and spray drying as a final step in the extraction (Muhoza et al., 2024; Rivera del

Rio et al., 2020; Yang et al., 2022).

3.1.3. Proteome profile of pea ingredients

Bottom-up proteomics is a commonly used analytical procedure in which proteins are enzymatically hydrolyzed and the resulting peptides are analyzed and further identified (Duong & Lee, 2023). Proteomics was used to track the effect of thermal treatment on protein identification and modifications of pea ingredients (Fig. 2). The abundance of the

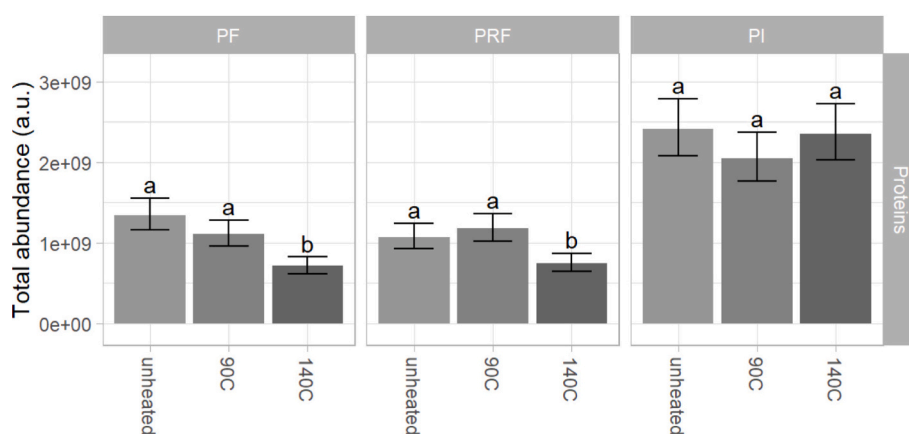


Fig. 2. Total abundance of proteins (arbitrary numbers) identified by LC-MS/MS in pea flour (PF), protein-rich fraction (PRF), and protein isolate (PI). Results are given as estimated means with 95% confidence intervals. Comparisons were done by Tukey grouping across the 3 heating treatments within pea sample, where heat treatments not sharing a letter are significantly different at the 5 pct significance level.

protein was estimated based on the MS signal from the individual peptides that were identified. The total abundance of proteins in PF and pea PRF heated at 140 °C was slightly lower than in the corresponding samples heated at 90 °C and unheated. Conversely, heating did not affect the total abundance of proteins in the pea PI. The low abundance of proteins identified in the PF and PRF could be a result of poor tryptic digestion of these ingredients, since the protein concentration was normalized for all samples. In these two samples, the presence of other components, such as fiber and starch, as well as protease inhibitors, can hamper tryptic digestion (Li et al., 2024; Vogelsang-O'Dwyer et al., 2020). On the other hand, the lower abundance of proteins seen in samples heated at 140 °C might be explained by protein aggregation and crosslinking, which limit the access of the trypsin to the proteolytic sites in the proteins. Surprisingly, the pea PI sample heated at 140 °C did not show a reduced abundance of proteins compared to what was observed in PF and PRF, despite the indication of protein aggregates seen in Fig. 1B.

3.1.4. Protein oxidation in pea ingredients

The total abundance of modified peptides by the oxidation of methionine and lysine is shown in Fig. 3. In this study, only oxidation products of methionine and lysine were investigated, since sulfur-containing amino acids and lysine belong to the amino acids most prone to be oxidized (Heinonen et al., 2021). One of the major carbonyl products of radical-induced oxidation is the α -aminoaldehyde (AAS) from lysine oxidation, which has been reported as a protein oxidation marker (Heinonen et al., 2021).

Methionine monooxidation was slightly higher in PF heated at 140 °C compared to the other conditions for PF, and lower in the PRF and PI at 140 °C, compared to their unheated and heated at 90 °C

counterparts. The observed reduction in methionine monooxidation following thermal treatment at 140 °C does not indicate a decrease in overall protein oxidation. Instead, it could be a result of further oxidation of methionine into methionine sulfone (dioxidation), as evidenced in the PI. Methionine undergoes rapid oxidation into methionine sulfide (monooxidation) under processing conditions involving reducing sugars, reductones, and Amadori products, subsequently leading to the formation of methionine sulfone (dioxidation) (Hellwig, 2019).

The effect of heat treatment at 140 °C on the oxidation of lysine was pronounced in the pea PRF and PI, while not observed in the PF. The identification of oxidized amino acids might be affected by low solubility and protein aggregation observed in oxidized proteins (Duque Estrada et al., 2018; Wang et al., 2025). AAS was observed in peptides originating from vicilin and some metabolic enzymes, such as lip-oxygenase, while the oxidation of methionine was identified in peptides originating from legumins and albumins. The oxidation of methionine observed in legumins can be explained by their composition, as legumins contain more sulfur-containing amino acids than vicilin. Zhang et al. (2024) observed that the legumin fraction in pea protein concentrate low-moisture extrudates was the most prone to oxidation using LC-MS/MS, with monooxidation being the most common type of modification. Methionine and lysine, as well as cysteine and arginine, are the most susceptible amino acids to oxidation, as also identified by proteomics in soybean meal subjected to storage at 35 °C (Wang et al., 2025).

3.2. Effect of thermal treatment on nutritional properties

3.2.1. Protease inhibitors

Fractionation methods and thermal treatment are usually efficient in removing and/or inactivating protease inhibitors in foods (Amin et al.,

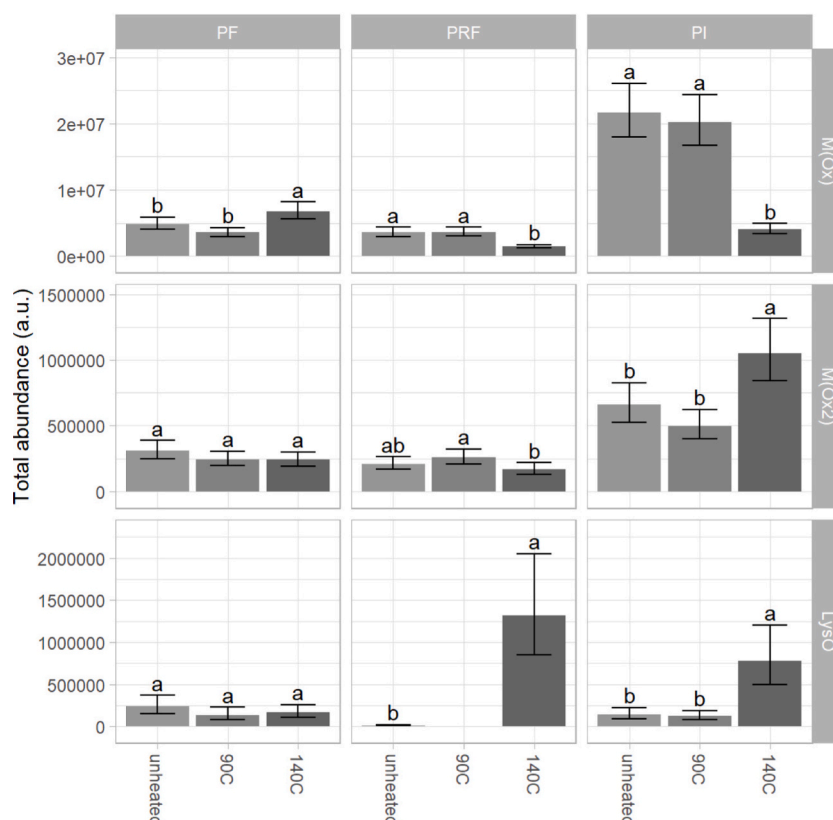


Fig. 3. Total abundance (arbitrary numbers) of peptides modified by oxidation of methionine (Met) and lysine (Lys) identified by LC-MS/MS, in pea flour (PF), protein-rich fraction (PRF), and protein isolate (PI) unheated and heated samples at 90 °C and 140 °C. Results are given as estimated means with 95% confidence intervals. Comparisons were done by Tukey grouping across the 3 heating treatments within the nine combinations of pea sample and protein, where heat treatments not sharing a letter are significantly different at the 5 pct significance level. M(Ox): methionine monooxidation; M(Ox2): methionine dioxidation; LysO: α -aminoaldehyde from lysine oxidation.

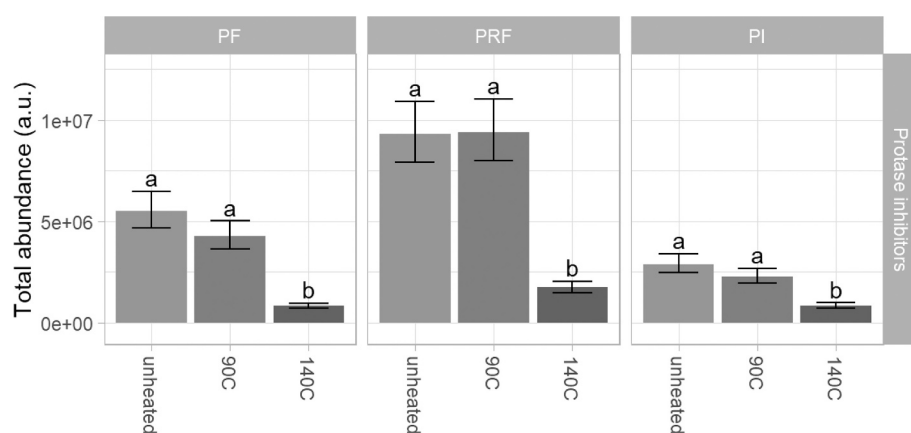


Fig. 4. Total abundance of protease inhibitors identified by LC-MS/MS in pea flour (PF), protein-rich fraction (PRF), and protein isolate (PI) before and after thermal treatment at 90 °C and 140 °C. Results are given as estimated means with 95% confidence intervals. Comparisons are done by Tukey grouping across the 3 heating treatments within pea sample, where heat treatments not sharing a letter are significantly different at the 5 pct significance level.

2022; Kong et al., 2022). For this reason, we investigated the impact of thermal treatment on the identification of the protease inhibitors, BBI and KTI, using proteomics (Fig. 4), as well as on the degree of inactivation of trypsin inhibitor activity in pea ingredients (Table 2).

Protease inhibitors were identified in the pea ingredients by proteomics, as shown in Fig. 4. The effect of heat treatment, at 140 °C, on the total abundance of protease inhibitors was pronounced in the pea PRF, showing that the dry fractionation concentrated the inhibitors in this fraction. Compared to wet fractionation, dry fractionation has been described to concentrate protease inhibitors in different sources of plant proteins alongside with the remaining protein fraction (Manzanilla-Valdez et al., 2024). PI showed the lowest total abundance of protein inhibitors among all samples, aligning with expectations that these compounds are typically low in protein isolates, rich in globulins. Protease inhibitors are water-soluble, low molecular weight proteins that belong to the albumin fraction; their solubility in the extraction medium facilitates their removal during the isoelectric precipitation step of the wet fractionation (Amin et al., 2022).

Regarding the effect of thermal treatment, heating at 90 °C did not change the total abundance of protease inhibitors compared to the unheated samples. However, heating at 140 °C resulted in a drastic reduction of inhibitors identified. At 140 °C, proteins were aggregated, resulting in less tryptic digestion and, therefore, a lower identification of peptides. These results are in accordance with the SDS-PAGE (Fig. 1B), where protein aggregation was seen in all samples heated at 140 °C.

Table 2 shows the effect of fractionation and thermal treatment on the TIA of pea ingredients. Unheated PF and PRF had higher TIA compared to PI. The lower TIA for PI was expected since the wet fractionation process is well known to remove most of the TIs from the protein ingredients (Amin et al., 2022). In addition, the drying step in

the wet fractionation is also expected to reduce the remaining TIA in protein isolates (Manzanilla-Valdez et al., 2024). Vogelsang-O'Dwyer et al. (2020) found a similar pattern of TIA being reduced in a faba bean protein isolate compared to faba bean flour and protein-rich fraction, showing one advantage of the wet fractionation compared to the dry fractionation on reducing the TIA and possibly resulting in improved protein digestibility.

Thermal treatment at 140 °C was more efficient in reducing TIA across all samples compared to the 90 °C treatment, except in the PF. Although the TIA levels of PF appear slightly higher at 140 °C compared to 90 °C, this difference may be attributed to methodological variability. It is plausible that thermal inactivation reaches a plateau at 90 °C, limiting further reduction in TIA levels despite higher processing temperatures. In the PI, the effect of the thermal treatment was less pronounced compared to the other samples, which aligns with the low TIA observed in the unheated sample. Xu et al. (2012) evaluated the correlation between TIA loss and heat-induced protein aggregation present in the supernatant and precipitate of ultracentrifuged soymilk. The authors showed that heating soymilk at 100 °C for 15 min reduced its TIA to about 20% of the raw level. This treatment also produced more precipitate than heating at 60–90 °C. Of the residual TIA, more than 80% remained in the supernatant, while the rest was found in the floating and precipitate fractions. The results showed a strong correlation between the amount of precipitate and TIA loss, which was explained by the incorporation of protease inhibitors into small and large protein aggregates precipitated at high temperatures.

3.2.2. *In vitro* protein digestibility of pea ingredients

The pea ingredients were digested *in vitro* for 2 h in the gastric phase and 2 h in the intestinal phase, using the INFOGEST method. The release

Table 2
Trypsin inhibitor activity of unheated and heated pea ingredients.

Sample	Thermal treatment	TUI per mg protein	Reduction
Pea flour	Unheated	18.89 [18.09; 19.73] a	
	90 °C	3.47 [3.31; 3.62] e	82% [80%; 83%]
	140 °C	4.35 [4.16; 4.54] d	77% [75%; 79%]
Protein rich-fraction	Unheated	18.18 [17.41; 18.99] a	
	90 °C	5.16 [4.94; 5.39] c	72% [69%; 74%]
	140 °C	2.40 [2.29; 2.51] f	87% [85%; 88%]
Protein isolate	Unheated	6.57 [6.28; 6.86] b	
	90 °C	4.64 [4.44; 4.85] d	29% [24%; 35%]
	140 °C	2.01 [1.92; 2.10] g	69% [67%; 72%]

Results are shown as estimates with 95% confidence intervals. TUI indicates *Trypsin Units Inhibited* per mg of protein (dry basis). Reduction indicates *relative decrease in TUI after thermal treatment*. Comparisons are done by Tukey grouping across the nine combinations of sample and thermal treatment, where combinations not sharing a letter are significantly different at the 5 pct significance level.

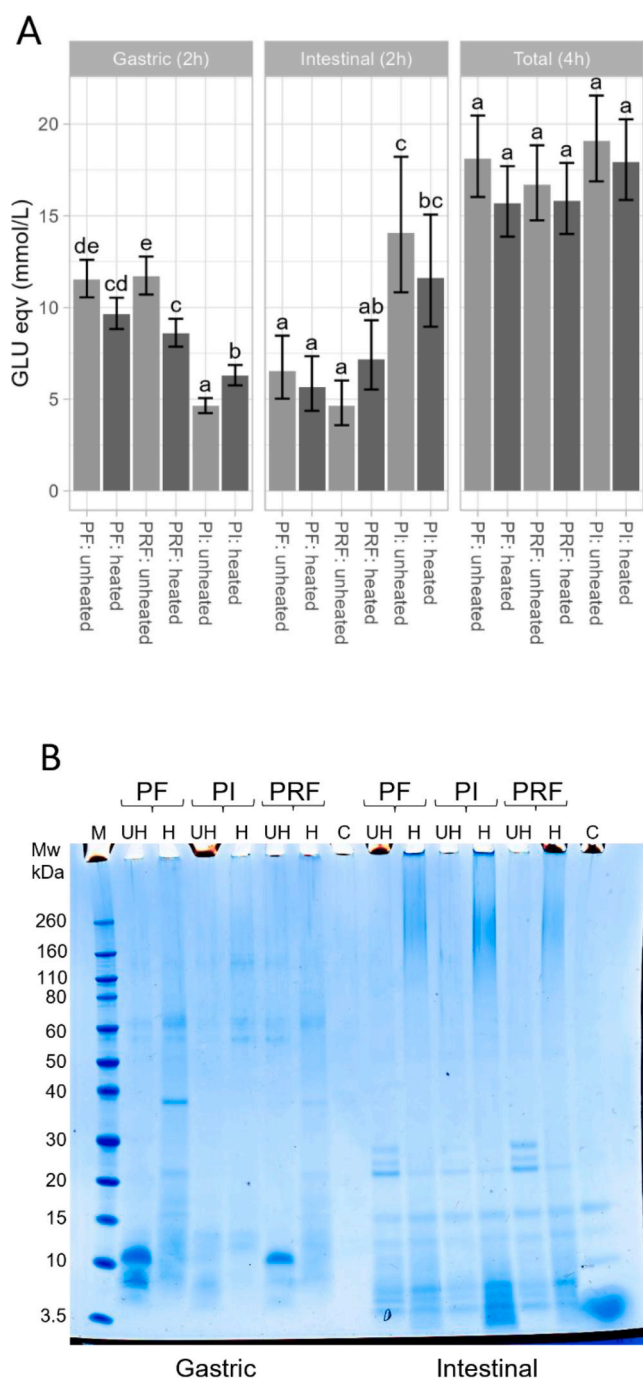


Fig. 5. A: Release of free amino groups during *in vitro* protein digestion shown as the concentration of glutamic acid equivalent (GLU eqv) after the gastric phase (2 h), intestinal phase (2 h), and in total (4 h). Results are shown as estimated marginal means with 95 pct confidence intervals in the error bars. Comparisons are done by Tukey grouping within the three digestion phases, where combinations of pea sample and heat treatment not sharing a letter are significantly different at the 5 pct significance level. B: Molecular weight profile of samples after the gastric and intestinal phase obtained by SDS-PAGE, under reducing conditions. UH: unheated, H: heated at 140 °C. M: molecular marker. C: protein-free cookie blank containing digestive enzymes. PF: pea flour. PI: protein isolate. PRF: protein-rich fraction.

of free amino groups during each phase of the *in vitro* protein digestion of unheated and heated samples at 140 °C is shown in Fig. 5A. In the gastric phase, the hydrolysis patterns of PF and PRF were similar, with both samples exhibiting lower hydrolysis when heated. When looking only at

Table 3

Estimates and 95 pct confidence intervals for the ratio between heated and unheated samples for the different pea ingredients in the gastric and the intestinal phase.

Phase	Sample	Ratio	p-value	Adjusted p-value
Gastric	PF	0.84 [0.74; 0.95]	0.0060	0.0240
Gastric	PRF	0.74 [0.65; 0.83]	0.0001	0.0001
Gastric	PI	1.36 [1.20; 1.54]	0.0001	0.0001
Intestinal	PF	0.87 [0.60; 1.25]	0.4381	0.6018
Intestinal	PRF	1.55 [1.07; 2.23]	0.0221	0.0661
Intestinal	PI	0.83 [0.57; 1.20]	0.3009	0.6018

The p-values are for the hypothesis of ratio equal to 1, and the adjusted p-values are computed using the Holm correction for the family-wise-error-rate across the six combinations of phase and sample. PF: pea flour; PRF: protein-rich fraction; PI: protein isolate.

the effect of thermal treatment on gastric protein digestion within the same group type (Table 3), a negative effect of heating was observed on PF, with a reduction of 16%, and on PRF, with a reduction of 26%. In addition to its effects on proteins, heating induces starch gelatinization, which may hinder protein digestibility, as observed during gastric digestion of quinoa protein-rich fraction and protein isolate mixed with starch heated at 120 °C (Opazo-Navarrete et al., 2019). Similarly, Oñate Narciso and Brennan (2018) observed a decrease in *in vitro* protein digestibility in mixtures of pea protein and starch, explained by the integration of proteins into the amylose network formed during gelatinization and cooling.

Here, the heated PI demonstrated improved hydrolysis in the gastric phase compared to the unheated sample, with a 35% increase (Table 3). This shows that heating contributed to better digestion of proteins under conditions where starch levels are low, despite the presence of heat-induced protein aggregates as demonstrated in the SDS-PAGE (Fig. 1B.1). In line with our findings, Tang et al. (2022) found that although thermal treatment at 65–100 °C for 30 min changed protein tertiary structure with the formation of new soluble protein aggregates, the degree of hydrolysis during the gastric phase increased compared to non-heated pea protein isolate, under semi-dynamic conditions simulating infant digestion. The authors attributed this enhanced digestibility to heat-induced protein unfolding, which increased surface hydrophobicity and thus improved accessibility to digestive enzymes.

Comparing across samples, both unheated and heated PF and PRF showed higher *in vitro* digestibility in the gastric phase than PI, regardless of heat treatment. This outcome is unexpected given the presence of starch and fiber in the PF and PRF matrices. These findings suggest that structural modifications introduced by wet fractionation, despite yielding a more purified protein matrix, actually limited gastric digestion. Subsequent heat treatment improved PI gastric digestibility, but not to a level exceeding that of PF and PRF. Another possible explanation for the higher protein digestion observed in the gastric phase for PF and PRF compared to the PI, may be associated with their higher content of free amino groups before digestion, as suggested by preliminary findings (data not shown). It is expected that the PI exhibit a low amount of free amino groups, as they are removed in the side-stream during the isoelectric precipitation of proteins. A higher initial content of free amino groups prior to the *in vitro* protein digestion analysis in the PF and PRF will contribute to the free amino groups detected post-gastric phase, independent of pepsin digestion. This could indicate that both PF and PRF are not better digested in the gastric phase compared to the PI. In support of this, Rivera del Rio et al. (2022) have shown a slightly higher proportion of peptides smaller than 0.5 kDa in pea flour than in pea protein isolate, both unheated. Unfortunately, the INFOGEST protocol is not designed to measure the concentration of free amino groups prior to digestion, and we chose not to introduce additional deviations from the standardized method. For this reason, the concentration of free amino groups before protein digestion was not assessed in the present study.

During the intestinal phase, an increase in protein digestion would have been expected at least for heated PF and PRF, due to the corresponding decrease in TIA (Fig. 5A). However, only the heated PRF showed higher protein digestion compared to the unheated sample, with an increase of 54% (Table 3). A previous study also reported improved *in vitro* protein digestibility in the intestinal phase after heating pea PRF at 90 °C, attributed to the inactivation of trypsin inhibitors (Rivera del Rio et al., 2022). Both the PF and PI showed lower protein digestion in the intestinal phase in the heated sample compared to the unheated sample, albeit neither of these was statistically significant.

Unheated PI had higher digestibility in the intestinal phase compared to PF and PRF, which can be explained by the low TIA observed in the PI (Table 2). Interestingly, once heated, PRF and PI showed no significant differences in digestibility in the intestinal phase. This suggests that the reduction in TIA in PRF, induced by thermal treatment, enabled it to achieve a similar degree of hydrolysis to PI. Despite the efficient reduction in TIA after heating PF, the same improvement in digestion in the intestinal phase was not observed. Probably the matrix composition plays a major role in the *in vitro* digestibility of PF due to its high content of starch and fiber, as discussed previously.

The total digestibility represents 4 h of protein digestion, combining the gastric and intestinal phases. Here, no differences were observed in the release of free amino groups between the heated and unheated samples, independently of the type of sample (Fig. 5A). Heating did not significantly increase the total protein digestion of PF and PRF, despite the inactivation of trypsin inhibitors (Table 2). Likewise, Hall and Moraru (2022) reported no effect on the *in vitro* protein digestion of heat-treated (95 °C for 15 min) pea protein concentrate obtained by air classification, despite a 70% reduction of trypsin inhibitor activity. The authors observed that protein aggregation was the predominant factor determining the protein digestibility of heat-treated pea protein concentrate compared to the unheated sample. Considering the impact of high temperatures combined with shearing, as occurs during high-moisture extrusion, a study using a broad temperature range (100–160 °C) has also shown no significant effect on the final *in vitro* protein digestibility of soy protein concentrate and isolate (de Boer et al., 2025). When considering only the final protein digestibility value after 4 h, no clear effect of thermal treatment or fractionation route is observed in the pea ingredients, despite the high heating conditions applied (140 °C for 20 min). However, when the digestion process is evaluated at each digestion phase, the impact of thermal treatment becomes evident. This highlights the importance of examining digestibility across all stages rather than relying solely on this 4 h endpoint, which obscures important differences. For future studies, employing semi-dynamic digestion models would be particularly valuable, as they can provide deeper insight into how thermal treatment affects digestion kinetics over time.

Fig. 5B shows the molecular profile of proteins after protein digestion in the gastric and intestinal phases. Low molecular weight proteins are seen after the gastric phase, as expected. It seems that more proteins are detected in the heated PF, between 30 and 15 kDa, after the gastric phase, indicating lower digestibility at this stage, as shown in Fig. 5A. These results align with the reduction in gastric digestion observed in heated PF, as presented in Table 3. However, it is difficult to draw definitive conclusions based on the SDS-PAGE, as these lanes appear smeared. The band observed around 40 kDa is probably pepsin, as shown by (Zhang et al., 2023).

After the intestinal phase, bands around 30 kDa appear in the unheated PF and PRF samples, which were not observed after the gastric phase. This could indicate a preservation of the trypsin and chymotrypsin from autolysis, in the presence of protease inhibitors, in comparison to samples that contained lower levels of the inhibitors (Table 2). The other bands seen in the intestinal phase at low molecular weight originate from the other enzymes present in the pancreatin solution, as can be seen in the protein-free cookie blank.

3.3. Proteomics as a complementary approach to evaluate plant protein nutritional quality

Given that plant protein ingredients produced through different fractionation routes differ in matrix composition and nutritional quality, assessing only *in vitro* protein digestibility does not fully capture the impact of processing on their overall quality. In complex matrices, components such as starch, fibers, and antinutrients, including protease inhibitors, can mask or counteract digestion differences. Moreover, thermal treatment can induce protein aggregation and amino acid modifications, for example due to oxidation, which may not be evident when looking solely at total digestibility. For this reason, we applied proteomics as an additional tool to identify heat-induced changes in the identification of proteins and oxidation of peptides. Unfortunately, differences in matrix composition affected the samples' behavior during tryptic digestion prior to LC-MS/MS, which limited direct comparisons as explained before. Nevertheless, we were still able to interpret the results by considering the effects of thermal treatment on each individual pea ingredient. Despite protein oxidation and protein aggregation (Fig. 1.B1) being more pronounced in the PI, its final *in vitro* protein digestibility was similar to that of the other ingredients. Because the oxidation values we report are relative rather than absolute, it is possible that the observed increase in oxidation reflects changes occurring in a small fraction of the total protein. This may help explain why the higher degree of oxidation and aggregation in the PI did not translate into lower overall digestibility. In addition, the lower level of TIA in the PI may have offset the potential negative effects of protein oxidation and aggregation. Considering the limitations of a static *in vitro* digestion model, *in vivo* studies would be relevant to better understand the impact of these modifications on the bioavailability of indispensable amino acids, and whether significant levels of modified amino acids are present in commercially available plant-based food products.

4. Conclusions

This study demonstrates how thermal processing influences the protein profile and *in vitro* protein digestibility of pea ingredients produced through wet and dry fractionation. The observed differences in protein profile and *in vitro* digestibility among PF, PRF, and PI are linked to variations in their matrix composition, extent of protein aggregation, TIA level, and the specific processing steps involved in producing each ingredient. These differences also affected digestive behavior, especially during the gastric phase in all samples, and only for PRF in the intestinal phase. However, at 140 °C, all samples showed extensive aggregation, and the inactivation of trypsin inhibitors helped counterbalance the matrix-related differences. Consequently, after 4 h of digestion, neither sample type nor heat treatment had a measurable effect on overall digestibility. From a nutritional perspective, and acknowledging the limitations of *in vitro* digestion models, the main implications of high-temperature treatment relate to changes in protein profiles and increased oxidation. These effects are more likely to influence functional properties, such as solubility and texturization, than overall protein digestibility. Further research is needed to clarify how these heat-induced modifications affect amino acid bioavailability and gut health. Additional studies, including complementary *in vitro* models and assessments of functional performance in real food applications, would help confirm these findings and determine their relevance for product development.

CRedit authorship contribution statement

Patrícia Duque-Estrada: Writing – original draft, Visualization, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Bo Markussen:** Writing – review & editing, Visualization, Formal analysis. **Maarten Schutyser:** Writing – review & editing. **Atze Jan van der Goot:** Writing – review &

editing. **Jens Christian Sørensen**: Writing – review & editing. **Keld Ejdrup Markedal**: Writing – review & editing. **René Lametsch**: Writing – review & editing, Formal analysis, Conceptualization. **Iben Lykke Petersen**: Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used Microsoft Copilot and Grammarly to assist with language editing in some parts of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodchem.2026.149181>.

Data availability

Data will be made available on request.

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